THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 253

THE TORPEDO POD

THE ARK NEVER DIVERTS

rescue or supplies for those we cannot turn for

side-mounted — 5g normal to the human

the lunar throw at 2.4 km/s — the Moon, first service station

V1 LANE FERRY · V2 ORBIT TARGET · NO ROOF TREBUCHET

THE POD GOES — THE SHIP HOLDS COURSE

Rescue logistics — v1.0 in force

GP-IM-253

Revision v1.0 — 24 August 2026

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VETTED BATCH 36 — TEST · STRUCTURAL AND HUMAN FACTORS

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THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 253

PHASE 253: THE TORPEDO POD — STATUS: CURRENT — TEST CLASS / STRUCTURAL + HUMAN FACTORS + TRAJECTORY VALIDATION (VET BATCH 36). No extra single-level roof trebuchet on the roof at rear — the torpedo pod, with man, oxygen and food, sent as a rescue mission or full of requested supplies to someone we cannot turn around or off course for. The third never-divert law: the ark never diverts — not for logistics (251's mules run the ferry), not for reconnaissance (252's salvo), and now not for rescue: the pod goes, the ship holds course. Side-mounted so the g-force is normal to a human at 5g — the safety line, graded: correct as a PEAK for short burns, Apollo/Soyuz-class territory. The sling is a throw, not a gun: at the crewed 5g cap a ring gives v = root(a.r). Consumables about 1 kg of oxygen and 2 kg of food per man-day — a two-week pod carries about 100 kg. The lane envelope is huge: matching a 10,000 mph relative drift takes 91 seconds at 5g. THE LUNAR THROW, his spec: the launcher must throw the pod off the Moon — 2.4 km/s required — and the Moon becomes the Highway's first service station: shallow well, no air, free rocks for the bolo. V1 the lane ferry; V2 the orbit target, his pickup, seated for version 2 — brake into the field.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-253, v1.0 — 24 August 2026. The two hundred and fifty-fourth manual of the program — 254 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the pod law as seated (contents line, the design as spoken, the third never-divert law, the launcher physics, the pod and the man, the bench items) — §2 the physics, graded — §3 the system in the pod — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 253: **The Torpedo Pod (The Courier Torpedo — Rescue or Supplies for Those We Cannot Turn For; the Ark Never Diverts).** Single-level roof trebuchet at the rear roof (229's sling family) throwing a torpedo pod — man, oxygen, food, or requested supplies; side-mounted so g rides transverse at the 5g safety limit (peak, short burns; cruise 1-2g); soft sling for the man, hard sling for the cargo; lane-rescue round trips at 10-18 km/s Δv; V1 the lane ferry, V2 the orbit target (well-rated, surface excluded by physics); the lunar throw: 2.4 km/s release off the Moon (10 km rail + coffee-cup pod burn, or the 251-family rotavator).'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

THE DESIGN, AS SPOKEN (VERBATIM)

Archer: "no extra single level roof trebuchet on the roof at rear, torpedo pod with man oxygen and food, being sent as a rescue mission or just full of requested supplies to someone we cannot turn around or off course for, side mounting the pod so g force is normal to a human at 5g, the safety limit"

Archer: "for gravitational field i was thinking maybe even an orbit target, not everyone has a ship always up from a planet not even earth, nice pickup for V2 though"

Archer: "ok closer to home based on that 1969 tin can weighing over a ton and a coffee cup full of fuel, this must be able to throw this pod off the moon"

THE DOCTRINE — THE THIRD NEVER-DIVERT LAW (VERBATIM)

- The ark never diverts — not for logistics (251's mules run the ferry), not for reconnaissance (252's salvo flies the look), and now not for people: **the torpedo goes instead.** Five hundred thousand tons does not maneuver for one man or one crate — the message is fired, not flown to. That is not cold; it is the only version of rescue that exists at fleet mass.
- **The boundary, seated so no reader overclaims:** a stranded giant belongs to 251's bolo; the far fast look belongs to 252's salvo; the pod ferries the near and the breathing. Three doctrines, one dovetail.

THE LAUNCHER — THE PHYSICS NOTE (VERBATIM)

- **The sling is a throw, not a gun:** at the crewed 5g cap, a ring gives $v = \sqrt{(a \cdot r)}$ — 10 m/s on 2 meters, 31 m/s on a 20-meter arm. The trebuchet's real jobs: clear the hull, orient the pod, and pay the first meters so the pod's own drive pays less. Two launch profiles, one launcher: **soft sling for the man (5g cap), hard sling for the cargo** — supplies feel nothing, so the same trebuchet can rip a cargo pod out at real speed and the pod rocket only finishes.
- **Consumables:** ~1 kg of oxygen and ~2 kg of food per man-day — a two-week pod carries ~100 kg of life. Nothing there.

- **The lane envelope is huge:** matching a 10,000 mph relative drift takes 91 seconds at 5g; a full lane-rescue round trip (go, stop, dock, come home) runs 10–18 km/s of Δv — a mass ratio under 1.5 on mule exhaust. The pod catches anything in the lane family, docks, and comes home.
- **THE LUNAR THROW (his spec, seated): the launcher must throw the pod off the Moon — 2.4 km/s release.** His benchmark honored: the 1969 tin can's ascent stage needed ~2 km/s of Δv on ~2.4 tonnes of propellant — not quite a coffee cup, but the Moon's well holds 1/22 of Earth's grip per kilogram, so the cup is directionally true. Three ways: (1) cargo rail — under 3 km of track at 100g, pocket change; (2) the crewed sweet spot — 10 km of rail at 5g for 1 km/s, the pod's rocket covers the remaining 1.4 km/s at a mass ratio of 1.03: the coffee cup, literally; (3) the rotavator — ~118 km arms at 5g crew comfort, 56% of the 20 GPa yarn ceiling, built from 251's own tether family — and it catches as well as it throws, so the sling runs on bookkeeping.
- **The Moon becomes the Highway's first service station:** shallow well, no air, free rocks for 251's stockpile, and a thrower that flings pods at escape speed for the price of electricity. The first bolo stockpile and the first pod launcher want the same address.

THE POD AND THE MAN (VERBATIM)

- **The 5g call, graded:** correct as a PEAK for short burns — Apollo/Soyuz-class territory — and the side-mounting is exactly right: g through the chest (eyeballs-in) is tolerable where g through the spine blacks you out. Clarification seated: 5g is minutes-class; the pod's working profile is 1–2g sustained, which a crew tolerates for hours to days, with 5g reserved for the punch burns.
- **V1 — THE LANE FERRY:** soft sling, pod burn, 1–2g cruise with 5g punch, weeks of air and food; catches anything in the lane family.
- **V2 — THE ORBIT TARGET (his pickup, seated for version 2):** well-rated — brake into the field, match orbit (~7.8 km/s around an Earth-class world), grapple, climb back out: ~20–30 km/s of budget, roughly triple V1, still inside mule-exhaust ratios; aerobrake where there is sky (the 252 shield-nose's cousin). **Surface excluded by physics, stated so no one oversells:** an Earth-class surface needs ~9 km/s of staged chemical climb — a lander, not a torpedo. V2's honest spec: rendezvous with anyone who can reach orbit — a station, a crippled ship, an escape pod — not anyone who can only reach the roof. His case recorded: 'not everyone has a ship always up from a planet not even earth' — Earth itself spent 2011–2020 unable to launch its own people on demand; if a planet-spanning civilization can have a crewed-launch gap, anyone can. The ark never diverts, and the well never gets a vote.

BENCH ITEMS — PHASE 253 (VERBATIM)

(1) The 5g transverse couch and restraint spec (minutes-class punch, eyeballs-in). (2) The consumables pack standard (per man-day, two-week module). (3) Lunar rail vs rotavator trade study for the first

service station. (4) Guidance — 81's intercept matrix heritage: the pod aims at moving targets by birthright. (5) Cargo-pod hardening standard for the hard sling.

ORIGIN OF THOUGHT — PHASE 253 (VERBATIM)

Born from the 1-g game — 21 mph per second — and from a 1969 tin can that proved how shallow the Moon's well is. The lineage: 229's roof sling is the mother (a single-level trebuchet, rear roof, dedicated), 41's mules lend the drive, 252's salvo lends the shield-nose cousin, 251 lends the tether family for the rotavator option and the never-divert doctrine itself, 81 lends the aim. Locked by number on the author's word: '253'.

Build the Torpedo Pod: the last-resort battery — a pod of torpedoes for the situation nothing else resolves. Carried and prayed over in the hope it stays sealed. — INSTRUCTION RECONSTRUCTION (NOT VERBATIM — the original order was not preserved; reconstructed from the seated body, 10 August 2026, per sitting 146).

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE THIRD NEVER-DIVERT LAW, SEATED: the ark never diverts — not for logistics (251's mules run the ferry), not for reconnaissance (252's salvo), and now not for rescue. The torpedo pod is the third leg: someone we cannot turn around or off course for still gets the man, the oxygen, the food — the pod goes, the ship holds course. The boundary is seated so no reader overclaims: a stranded giant belongs to 251's bolo; the pod carries the small and the urgent.

THE LAUNCHER — THE SLING IS A THROW, NOT A GUN: at the crewed 5g cap, a ring gives v = root(a.r) — 10 m/s on a small ring, scaling with radius; the roof-trebuchet amendment is his own kill: 'no extra single level roof trebuchet on the roof at rear' — the pod rides the existing throw machinery, side-mounted, and the launcher question is the phase's physics note.

THE 5G CALL, GRADED: correct as a PEAK for short burns — Apollo/Soyuz-class territory — with the g normal to the body by the side-mount. The human-factors item the vet named is the long-burn posture, and the phase owes it.

THE CONSUMABLES, PRICED: about 1 kg of oxygen and 2 kg of food per man-day — a two-week pod carries about 100 kg. The lane envelope is huge: matching a 10,000 mph relative drift takes 91 seconds at 5g — the pod is a ferry, not a chase vehicle, and the numbers say the ferry covers the lane.

THE LUNAR THROW — HIS SPEC, SEATED: the launcher must throw the pod off the Moon — 2.4 km/s required. The Moon becomes the Highway's first service station: shallow well, no air, free rocks for the bolo line, and a throw that needs no atmosphere pricing. The shakedown's day trip is its first customer.

V1 AND V2, SEATED IN ORDER: V1 — THE LANE FERRY: soft sling, pod burn, 1-2g cruise with the 5g punch in reserve, weeks of air and food. V2 — THE ORBIT TARGET (his pickup, seated for version 2): well-rated — brake into the field. The version ladder is the doctrine: fly the ferry first, then rate the deep catch.

§3 — THE SYSTEM IN THE POD

- **The law:** the ark never diverts — the third never-divert; the pod goes, the ship holds course.
- **The pod:** man, oxygen, food — about 100 kg for two weeks; rescue or requested supplies.
- **The mount:** side-mounted, g normal to the human, 5g peak cap — Apollo/Soyuz-class territory.
- **The throw:** the sling is a throw, not a gun — v = root(a.r) at the 5g cap; no roof trebuchet, his kill.
- **The envelope:** 91 seconds at 5g matches a 10,000 mph drift — the ferry covers the lane.
- **The Moon:** 2.4 km/s off the lunar well — the Highway's first service station.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- The never-divert laws now stand at three: 251's mules for logistics, 252's salvo for reconnaissance, 253's pod for rescue — the ark's course is the law, and the exceptions are machines, not turns.
- 251's bolo owns the boundary: a stranded giant belongs to the tow truck; the pod's jurisdiction is the small and the urgent — seated so no reader overclaims either machine.
- 244's shakedown is the first customer: the Moon day trip puts the pod's launcher where the lunar throw is the day's work, and the service-station law gets its opening shift.
- The roof-trebuchet kill is his own amendment, seated verbatim: 'no extra single level roof trebuchet on the roof at rear' — the design deleted its own launcher before the audit could ask.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 36 (17 August 2026 — phases 251-255 plus the 251 bolo rebuttal, cross-checked in sitting 405), SEATED. The grade, verbatim from the batch: '253 → TEST / STRUCTURAL + HUMAN FACTORS + trajectory validation.' The remaining work named by the verdict itself: structural, human-factors and trajectory validation.

§6 — BUILD SEQUENCE

- 1. Rate the launcher: v = root(a.r) at the 5g cap — ring radius versus throw speed, the lunar 2.4 km/s spec benched.

- 2. Fit the mount: side-mount, g normal to the human — the short-burn posture proven, the long-burn owed.
- 3. Stock the pod: about 1 kg oxygen and 2 kg food per man-day — the two-week 100 kg manifest sealed.
- 4. Fly the envelope: the 91-second 10,000 mph match demonstrated on the lane simulator, then on the lane.
- 5. Open the service station: the lunar throw as a shakedown task — the Highway's first service station opens with the day trip.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 251 (the bolo trail):** the boundary — the stranded giant is the bolo's job; the pod carries the small and the urgent; the never-divert laws divide the work.
- **Phase 252 (the encounter doctrine):** the sibling law — the salvo is the second never-divert; the pod is the third.
- **Phase 244 (the shakedown doctrine):** the first customer — the Moon day trip opens the service station; the lunar throw is the day's work.
- **Phase 229 (the sling):** the launcher family — the throw, not the gun; the pod rides the family's throw machinery at crewed caps.

§8 — DO-NOT-BUILD

- Do NOT divert the ark — the third never-divert law: the pod goes, the ship holds course; rescue is a machine, not a turn.
- Do NOT build the roof trebuchet — his kill, seated: no extra single-level roof trebuchet on the roof at rear.
- Do NOT exceed the 5g cap — peak, short burns, g normal to the body; the long-burn posture is owed, not assumed.
- Do NOT size it for giants — a stranded giant belongs to 251's bolo; the pod's jurisdiction is the small and the urgent.
- Do NOT skip the lunar rating — the launcher must throw 2.4 km/s off the Moon; the service-station law depends on it.

§9 — OWED

OWED: the vet's three, whole — structural validation of the pod and mount at the 5g peak; human-factors validation of the burn postures (short proven, long owed); trajectory validation across the lane envelope and the lunar 2.4 km/s throw. Plus the two-week consumables manifest benched, not summed.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-253, v1.0 — CURRENT — TEST CLASS / STRUCTURAL + HUMAN FACTORS + TRAJECTORY VALIDATION (VET BATCH 36), seated law, master v2.1 (numerical print order). The two hundred and fifty-fourth manual of the program — 254 of 290. Built 24 August 2026, fourteenth of the 20-manual 'ok next 20' shift (the author's order, 24 August 2026, seated in sitting 622). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618 and 622): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20'. The phase's own chain, all seated in the master: the contents line and the bodies (as seated — the 253 body carries the complete order in the seated 'Archer:' quotes) — the design, verbatim: 'no extra single level roof trebuchet on the roof at rear, torpedo pod with man oxygen and food, being sent as a rescue mission or just full of requested supplies to someone we cannot turn around or off course for'; the third never-divert law; the 5g side-mount; the lunar throw at 2.4 km/s and the Moon as the Highway's first service station; V1 the lane ferry, V2 the orbit target; the vet batch 36 grade (TEST / STRUCTURAL + HUMAN FACTORS + trajectory validation) in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the structural and human-factors bench and the lunar throw rating, or his word.

END OF MANUAL — PHASE 253